A4 with generative AI: designing your IIoT architecture poster in Claude

You may use a generative AI assistant to help make your A4 poster. This guide shows one way to do it with a free Claude account at claude.ai. It takes about three hours, most of it checking and editing. Claude can build an A1 PowerPoint slide in code, with the layers, colours and arrows already placed, which you then finish in PowerPoint and export to PDF. The marks for the protocol reasons, the outage strategy and the summary for management come from your own judgement about the Antananarivo baobab plant, so the guide keeps those parts in your hands.

Before you start

Sign in at claude.ai with a free account. Open Settings, then Capabilities, and turn on **Cloud code execution and file creation**. Claude needs it to read your PDFs and hand you a real `.pptx` file. You do not need to add any other skill: Claude's PowerPoint skill switches on by itself when you ask for a presentation file.

You also need PowerPoint (or LibreOffice Impress) to edit the slide and export the PDF.

If Claude stops and asks permission to run a command, click **Allow once**. The command runs in Claude's own sandbox in the cloud, not on your computer. A prompt that builds a file can take 5 to 15 minutes. Leave the tab open until Claude has finished and the file appears in the chat.

Start a new chat for this assignment and keep the whole job in that one chat, so Claude remembers what it has already built. A free account has a usage limit that resets every five hours. If you hit it, wait and carry on in the same chat.

Files to upload

Download these from the A4 assignment page on InnoSkills and attach all three to your first message, using the paperclip button:

1. `A04_iiot_architecture_poster.pdf`, your brief.
2. `A04_exemplar_lagos_cell_poster.pdf`, the worked exemplar poster.
3. `A04_exemplar_lagos_cell.pdf`, the marker's note on the exemplar.

The exemplar is set at a different site (a cobot inspection cell in Lagos), so its sensors, protocols and outage plan belong to a robot cell, not a drying tunnel. Use it to see the shape of a strong poster: one direction of flow, a colour and icon per layer, a legend, one signal traced from data to decision, a summary box for the plant manager, and no text smaller than 24 pt at print size.

Prompts

Copy each prompt into the chat in turn. Read the answer before you send the next one. Where a prompt has a square-bracketed gap, fill it in yourself.

1. Read the brief

```
I'm a learner on the DigiSkills programme. Attached are my A4 brief (PDF) and a worked exemplar poster plus its marker's note from a different site. Read the brief carefully. Then give me a checklist of every Key Requirement and every row of the marking scheme, with what a Distinction answer needs for each. Do not build anything yet. My poster is about the Ubuntu Naturals Antananarivo baobab primary processing plant and its Tier-3 sister plants, not the exemplar's Lagos cobot cell.
```

2. Choose the sensors

```
Propose a sensor table for baobab primary processing at Antananarivo, one group per step: drying, grinding, sieving, and ambient and storage. For each sensor give what it measures, a generic specification (range, accuracy, output signal), how often it is read and which layer it connects to. Do not name products or quote figures you cannot point to a source for. Mark any specification that is your estimate.
```

Decide which sensors to keep, and check every specification you keep against a datasheet or a lecture. Those choices are part of your mark.

3. Plan the layout

```
Plan a single A1 landscape poster (84.1 cm by 59.4 cm) for my design. My protocols are [field, edge-to-cloud and application choices] and my Industry 4.0 pillars are [at least four, with where each sits]. Lay out: a title block (plant name, date, my name), the five-layer architecture as the main panel with flow running one way, a data flow traced through the DIKW pyramid, a protocol panel, an outage panel, a pillar panel, a legend and a summary box for senior management. Give each panel its position, size and a word budget that keeps every piece of text at 24 pt or larger. Do not draft the protocol reasons, the outage panel or the summary box.
```

4. Build the poster

```
Using python-pptx, build A04_IIoT_Architecture_Poster_[YourSurname].pptx: one slide sized 84.1 cm by 59.4 cm, following the layout plan. Use native shapes, connectors and text boxes so I can edit everything in PowerPoint; do not flatten anything into a picture. Give each of the five layers its own colour and a simple icon, from a colour-blind safe palette that never relies on red against green. Number the steps of the traced signal. Put my sensor table in the sensor layer, showing each sensor, what it measures and its read interval, and put range, accuracy and signal in the speaker notes. Leave the protocol reasons, the outage panel and the summary box as text boxes that say [TO WRITE]. No text smaller than 24 pt. Follow the exemplar's layout ideas but none of its content or wording.
```

5. Check legibility and fit

```
With code, open the .pptx and list every text box with its smallest font size, and flag anything below 24 pt, any text box that overflows its shape and any shapes that overlap by accident. Then render the slide to a PNG with LibreOffice if you can and show it to me. Fix what you find and give me the corrected file.
```

6. Write the judgement panels yourself, then get feedback

Write the protocol reasons, the outage panel (buffering, what keeps running, recovery order, how buffered data is checked) and the summary box in your own words, for the Tier-3 family's power and

staffing constraints. Then paste each draft into the chat:

```
Here is my draft [protocol reasons / outage panel / summary box]. Mark it against the "[Sensor and Protocol Specification / Edge Computing and Resilience / Visual Design and Communication]" row of the marking scheme in my brief. Tell me which band it reaches and what would lift it one band. Tell me if it will fit its text box at 24 pt. Do not rewrite it for me. My draft: [paste your draft here]
```

When you are happy with a draft, type it into its text box in PowerPoint yourself. Then choose File, then Export, then PDF.

Check before you submit

- Open the PDF and confirm the page is A1 landscape, 841 mm by 594 mm.
- View the PDF at 100% zoom and read every panel. Nothing should be blurred, clipped or smaller than 24 pt.
- Print one panel on A4 at 100% scale (tile or crop) and read it from two metres away.
- Check the colours in greyscale: the layers must still be told apart, and red against green must never be the only difference.
- Confirm the title block carries the plant name, the date and your name, and that no [TO WRITE] box is left.
- Read the whole poster for leftover exemplar content: Lagos, cobot, AGV, lidar, NAFDAC or tray should not appear anywhere.
- Name the file `A04_IIoT_Architecture_Poster_[YourSurname].pdf` with your own surname.

Add a short line in the title block or a footnote saying which parts Claude built and which decisions and writing are yours. Claude can make mistakes. You are responsible for every number and every sentence in what you submit.